

Tugas Aljabar

Date.

No.

$$5x + 15y + 16z = 35$$

$$-4x - 11y - 11z = -26$$

$$-x - 3y - 11z = -7$$

Eliminasi variabel x pada pers 1 dan 2

$$\begin{array}{r|l} 5x + 15y + 16z = 35 & 4 \\ -4x - 11y - 11z = -26 & 5 \end{array} \quad \begin{array}{l} 20x + 60y + 224z = 140 \\ -20x - 55y - 205z = -130 \end{array} +$$

$$5y - 19z = 10 \rightarrow \text{Persamaan 4}$$

Eliminasi variabel x pada pers 1 dan 3

$$\begin{array}{r|l} 5x + 15y + 16z = 35 & 1 \\ -x - 3y - 11z = -7 & 5 \end{array} \quad \begin{array}{l} 5x + 15y + 16z = 35 \\ -5x - 15y - 55z = -35 \end{array} +$$

$$1z = 0$$

$$z = 0 \text{ Persamaan 5}$$

Eliminasi y pada pers 4 dan 5

$$5y - 19z = 10$$

$$5y = 10$$

$$5y - 19 \cdot 0 = 10$$

$$5y = 10$$

$$y = \frac{10}{5} = 2$$

Persamaan 3

$$-x - 3y - 11z = -7$$

$$-x - 3(2) - 11(0) = -7$$

$$-x - 6 = -7$$

$$-x = -7 + 6$$

Pasangan 1

$$5x + 15y + 56z = 15$$

$$5x + 15(4) + 56(0) = 35$$

$$5x + 30 + 0 = 35$$

$$5x + 30 = 35$$

$$5x = 35 - 30$$

$$5x = 5$$

$$x = \frac{5}{5} = 1$$

Eliminasi Gauss

$$5x + 15y + 56z = 35$$

$$-4x - 11y - 41z = -24$$

$$-x - 3y - 11z = -7$$

Konversi ke bentuk matriks

$$\begin{bmatrix} 5 & 15 & 56 \\ -4 & -11 & -41 \\ -1 & -3 & -11 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 35 \\ -24 \\ -7 \end{bmatrix}$$

Jalankan operasi OPE matrik Euklides sebagai atas

$$\begin{bmatrix} 5 & 15 & 56 \\ -4 & -11 & -41 \\ -1 & -3 & -11 \end{bmatrix} \begin{bmatrix} 35 \\ -24 \\ -7 \end{bmatrix} \Rightarrow H_{21}(4) = \begin{bmatrix} 1 & 3 & 56/5 \\ -4 & -11 & -41 \\ -1 & -3 & -11 \end{bmatrix} \begin{bmatrix} 7 \\ -24 \\ -7 \end{bmatrix} \Rightarrow H_{21}(4)$$

$$H_{21}(4) \begin{bmatrix} 1 & 3 & 56/5 \\ 0 & 1 & 12/5 \\ -1 & -3 & -11 \end{bmatrix} \begin{bmatrix} 7 \\ 2 \\ -7 \end{bmatrix} \Rightarrow H_{31}(1) \begin{bmatrix} 1 & 3 & 56/5 \\ 0 & 1 & 12/5 \\ 0 & 0 & -11 \end{bmatrix} \begin{bmatrix} 7 \\ 2 \\ 0 \end{bmatrix}$$

$$H_{32}(5) \begin{bmatrix} 1 & 3 & 56/5 \\ 0 & 1 & 12/5 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 7 \\ 2 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 3 & 56/5 \\ 0 & 1 & 12/5 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 7 \\ 2 \\ 0 \end{bmatrix}$$

Kembalikan ke bentuk persamaan

$$\begin{aligned} 5x + 15y + 56z &= 35 \\ -4x - 11y - 41z &= -24 \\ -x - 3y - 11z &= -7 \end{aligned} \Rightarrow \begin{aligned} x + 3y + 56/5z &= 7 \text{ Persamaan 1 ke 4} \\ y + 12/5z &= 2 \text{ Persamaan 1 ke 5} \\ z &= 0 \end{aligned}$$

Subst. hitungan 2 ke persamaan 5

Sub 2 dan y ke pers 4:

$$y + 14/5 z = 2$$

$$x + 3y + 56/5 z = 7$$

$$y + 12/5(0) = 2$$

$$x + 3(1) + 56/5(0) = 7$$

$$y = 2$$

$$x + 4 + 0 = 7$$

$$x + 4 = 7$$

$$x = 7 - 4$$

$$x = 1$$

Methode Invers

$$5x + 15y + 56z = 35$$

$$-4x - 11y - 41z = -26$$

$$-x - 3y - 11z = -7$$

$$\begin{bmatrix} 5 & 15 & 56 \\ -4 & -11 & -41 \\ -1 & -3 & -11 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 35 \\ -26 \\ -7 \end{bmatrix}$$

Carilah determinan dulu

$$A = \begin{bmatrix} 5 & 15 & 56 \\ -4 & -11 & -41 \\ -1 & -3 & -11 \end{bmatrix} \quad \det A = (5 \cdot (-11) \cdot (-11)) + (15 \cdot (-4) \cdot (-1)) + (56 \cdot (-4) \cdot (-1))$$

$$= (5 \cdot 121) + (15 \cdot 4) + (56 \cdot 4)$$

$$= (605 + 60 + 224)$$

$$= (605 + 60 + 224) = 889$$

$$= 889$$

$$\det A = 1$$

$$\det A = 1$$

$$\det A = 1$$

$$M_{11} = \begin{vmatrix} -11 & -41 \\ -3 & -11 \end{vmatrix} \quad M_{12} = \begin{vmatrix} -4 & -41 \\ -1 & -11 \end{vmatrix}$$

$$\det(M_{11}) = (-11) \cdot (-11) - (-3) \cdot (-41) = (-4) \cdot (-11) - (-1) \cdot (-41)$$

$$= 121 - 123 = 3$$

$$= -2$$

$$M_{13} = \begin{vmatrix} -4 & -11 \\ -1 & -3 \end{vmatrix} \quad M_{21} = \begin{vmatrix} 15 & 56 \\ -3 & -11 \end{vmatrix} \quad M_{22} = \begin{vmatrix} 5 & 56 \\ -1 & -11 \end{vmatrix}$$

$$= (-4 \cdot (-3)) - (-1 \cdot (-11)) = (15 \cdot (-11)) - (56 \cdot (-3)) = (5 \cdot (-11)) - (56 \cdot (-1))$$

$$= 12 - 11 = 1$$

$$M_{23} = \begin{vmatrix} 5 & 15 \\ -1 & -3 \end{vmatrix} \quad M_{31} = \begin{vmatrix} 15 & 56 \\ -11 & -41 \end{vmatrix} \quad M_{32} = \begin{vmatrix} 5 & 56 \\ -4 & -41 \end{vmatrix}$$

$$= (5 \cdot (-3)) - (-15 \cdot (-1)) = (15 \cdot (-41)) - (56 \cdot (-11)) = (5 \cdot (-41)) - (56 \cdot (-4))$$

$$= 80$$

$$= 1$$

$$19$$

$$M_{33} = \begin{vmatrix} 5 & 15 \\ -4 & -11 \end{vmatrix}$$

$$= (5 \cdot (-11)) - (-15 \cdot (-4))$$

$$= 5$$

Koraktor =

$$\begin{bmatrix} -2 & -3 & 1 \\ -3 & 1 & -19 \\ 1 & -19 & 5 \end{bmatrix}$$

$$T_{\text{koraktor}} = \begin{bmatrix} -2 & -3 & 1 \\ -3 & 1 & -19 \\ 1 & -19 & 5 \end{bmatrix}$$

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$$

 $\det(A)$

$$A^{-1} = \frac{1}{1} \begin{bmatrix} -2 & -3 & 1 \\ -3 & 1 & -19 \\ 1 & -19 & 5 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} -2 & -3 & 1 \\ -3 & 1 & -19 \\ 1 & -19 & 5 \end{bmatrix}$$

$$\begin{bmatrix} -2 & -3 & 1 \\ -3 & 1 & -19 \\ 1 & -19 & 5 \end{bmatrix}$$

$$X = (A^{-1})B$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -2 & -3 & 1 \\ -3 & 1 & -19 \\ 1 & -19 & 5 \end{bmatrix} \begin{bmatrix} 35 \\ -26 \\ -7 \end{bmatrix} \rightarrow \begin{bmatrix} (-2 \cdot 35) + (-3 \cdot -26) + (1 \cdot -7) \\ (-3 \cdot 35) + (1 \cdot -26) + (-19 \cdot -7) \\ (1 \cdot 35) + (-19 \cdot -26) + (5 \cdot -7) \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -70 + 78 + (-7) \\ -105 + (-26) + 131 \\ 35 + 490 + (-35) \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

Matode (cramer)

menentukan determinan tiap variabel

$$B A = \begin{bmatrix} 35 & 15 & 56 \\ -26 & -11 & -41 \\ -7 & -3 & -11 \end{bmatrix} \cdot \begin{bmatrix} 35 & 15 \\ -26 & -11 \\ -7 & -3 \end{bmatrix}$$

$$\begin{bmatrix} (35 \cdot (-11) \cdot (-11)) + (15 \cdot (-41) \cdot (-7)) + (56 \cdot (-26) \cdot (-3)) - ((15 \cdot (-26) \cdot (-11)) + (-26 \cdot (-11) \cdot (-3)) + (56 \cdot (-7) \cdot (-3)) \end{bmatrix}$$

$$= (41235 + 4305 + 4368) - (41290 + 4305 + 4312)$$

$$= 12908 - 12907$$

$$\det B = 1$$

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$$C = \begin{bmatrix} 5 & 35 & 56 \\ -4 & -26 & -41 \\ -1 & -7 & -11 \end{bmatrix} \quad \begin{bmatrix} 5 & 35 \\ -4 & -26 \\ -1 & -7 \end{bmatrix}$$

$$= ((5 \cdot (-26) \cdot (-11)) + (35 \cdot (-41) \cdot (-1)) + (56 \cdot (-4) \cdot (-7))) - ((35 \cdot (-4) \cdot (-11)) + (5 \cdot (-41) \cdot (-7)) + (56 \cdot (-26) \cdot (-1)))$$

$$= (1430 + 1435 + 1568) - (1540 + 1435 + 1456)$$

$$= 4433 - 4431$$

$$= 2$$

$$D = \begin{bmatrix} 5 & 15 & 35 \\ -4 & -11 & -26 \\ -1 & -3 & -7 \end{bmatrix} \quad \begin{bmatrix} 5 & 15 \\ -4 & -11 \\ -1 & -3 \end{bmatrix}$$

$$= (5 \cdot (-11) \cdot (-7) + (15 \cdot (-26) \cdot (-1)) + (35 \cdot (-4) \cdot (-3))) - (15 \cdot (-4) \cdot (-7)) + (5 \cdot (-26) \cdot (-3)) + (35 \cdot (-11) \cdot (-1))$$

$$= (385 + 390 + 420) - (420 + 390 + 385)$$

$$= 1195 - 1195$$

$$= 0$$

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$$\det A = 1$$

$$x = \frac{\det(x)}{\det(A)} = \frac{1}{1} = 1$$

$$\det B = 1$$

$$\frac{\det(A)}{\det(A)} = 1 //$$

$$\det y = 2$$

$$y = \frac{\det(y)}{\det(A)} = \frac{2}{1} = 2 //$$

$$\det z = 0$$

$$z = \frac{\det(z)}{\det(A)} = \frac{0}{1} = 0 //$$

Methode CSE

$$\begin{bmatrix} 5 & 15 & 56 & 75 \\ -4 & -11 & -41 & -26 \end{bmatrix}$$

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Methode OBE

$$= \begin{bmatrix} 5 & 15 & 56 & 35 \\ -4 & -11 & -26 & -26 \\ -1 & -3 & -11 & -7 \end{bmatrix} \xrightarrow{H_1(1/5)} \begin{bmatrix} 1 & 3 & 56/5 & 7 \\ -4 & -11 & -41 & -26 \\ -1 & -3 & -11 & -7 \end{bmatrix}$$

$$H_{21}(4) = \begin{bmatrix} 1 & 3 & 56/5 & 7 \\ 0 & 1 & 19/5 & 2 \\ -1 & -3 & -11 & -7 \end{bmatrix} \xrightarrow{H_{21}(1)} \begin{bmatrix} 1 & 3 & 56/5 & 7 \\ 0 & 1 & 19/5 & 2 \\ 0 & 0 & 1/5 & 0 \end{bmatrix}$$

$$H_{12}(-3) = \begin{bmatrix} 1 & 0 & 41/5 & 1 \\ 0 & 1 & 19/5 & 2 \\ 0 & 0 & 1/5 & 0 \end{bmatrix} \xrightarrow{H_{13}(1)} \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 19/5 & 2 \\ 0 & 0 & 1/5 & 0 \end{bmatrix}$$

$$H_{23}(-19) = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1/5 & 0 \end{bmatrix} \xrightarrow{H_3(5)} \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\begin{array}{l} x = 1 \\ y = 2 \end{array} \quad \Bigg| \quad z = 0$$